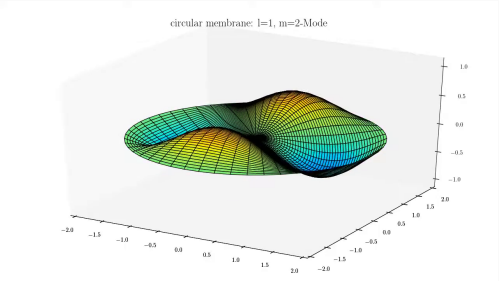
# Transient analysis

## Analysis of free and forced vibrations

ME473 Dynamic finite element analysis of structures

Stefano Burzio

2025



# Where do we stand?

| Week | Module | Lecture topic | Mini-projects |
|---|---|---|---|
| 1 | Linear elastodynamics | Strong and weak forms | |
| 2 | | Galerkin and FE methods | |
| 3 | | Systematisation of the procedure | Groups formation |
| 4 | | 3d elements, numerical integration | Project 1 statement |
| 5 | Special structural elements | Bars and trusses | |
| 6 | | Beams, frames and grids | |
| 7 | | Kirchhoff-Love plates | |
| 8 | | Reissner-Mindlin plates | Project 1 submission |
| 9 | | Shells | Project 2 statement |
| 10 | Free and forced vibrations | Analysis of free vibrations | |
| 11 | | Analysis of free vibrations | |
| 12 | | Transient analysis | Project 2 submission |

## Summary

- Recap week 11
- Analysis of forced vibrations
- Direct integration methods

## Recommended readings

(B) Bathe, Finite Element Procedures in Engineering Analysis (chap. 9)

(P) Petyt, Introduction to finite element vibration analysis (chap. 12)

(G) Gmür, Dynamique des structures (§5.1 and 5.2)

# General end-of-course information

# Final examination

- **Date:** January 14th
- **Location:** Room CM1221
- **Duration:** 20 min individual presentation followed by a 10 min discussion
- **Report:** Submit the project report via Moodle by January 9th

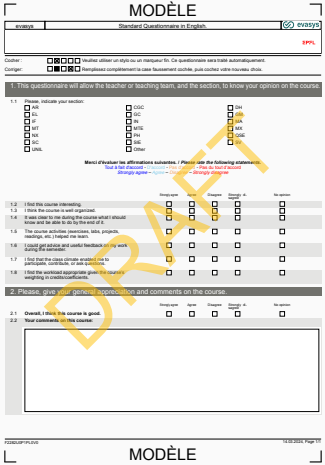



**Avoid unnecessary stress!**

**Preparation recommendations:**

- Review the lecture notes and relevant textbook chapters.
- Thoroughly review the assigned problem sets.
- Use Ed-discussion forum or drop by my office (ME A2 390) if you have any questions.

# CAPE evaluation survey


MODÈLE

evasys | Standard Questionnaire in English. | evasys

FPPL

Cocher : Veuillez utiliser un stylo ou un marqueur fin. Ce questionnaire sera traité automatiquement.

Corriger : Remplissez complètement la case fauxsement cochée, puis cochez votre nouveau choix.

1. This questionnaire will allow the teacher or teaching team, and the section, to know your opinion on the course.

1.1 Please, indicate your section:

NR | CGC | CH
EL | GC | DR
IF | IN | LA
MT | MTE | MX
NX | PH | QMC
SC | SIE | SV
VNXL | Other

Merci d'évaluer les affirmations suivantes. / Please rate the following statements:
Tout à fait d'accord – D'accord – Plutôt d'accord – Plutôt pas d'accord – Pas d'accord – Pas du tout d'accord
Strongly agree – Agree – Disagree – Strongly disagree

| | | Strongly agree | Agree | Disagree | Strongly di... | No opinion |
|---|---|---|---|---|---|---|
| 1.2 | I find this course interesting. | | | | | |
| 1.3 | I think the course is well organized. | | | | | |
| 1.4 | I know clearly the learning outcomes what I should know and be able to do by the end of it. | | | | | |
| 1.5 | The course activities (exercises, labs, projects, readings, etc.) helped me learn. | | | | | |
| 1.6 | I could get advice and useful feedback on my work during the semester. | | | | | |
| 1.7 | I find that the class climate enabled me to participate, contribute, or ask questions. | | | | | |
| 1.8 | I find the workload appropriate given the course's weighting in credits/coefficients. | | | | | |

2. Please, give your general appreciation and comments on the course.

| | | Strongly agree | Agree | Disagree | Strongly di... | No opinion |
|---|---|---|---|---|---|---|
| 2.1 | **Overall, I think this course is good.** | | | | | |
| 2.2 | **Your comments on the course:** | | | | | |

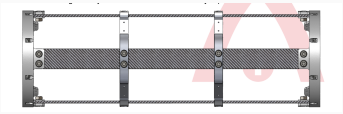
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MODÈLE

**Please complete the in-depth evaluation survey!**




**Student assistant (AE) needed !**

- We are seeking one, possibly two, teaching assistants (AEs) to support the course through:
  - Assistance with exercise sessions
  - Supervision of student mini-projects

**Semester projects**

- Structural and dynamic analysis via FEM
- Neural networks meet finite elements

**Master projects**

- In collaboration with D-Orbit (topic to be finalized)
- Experimental modal analysis

# Recap week 11
# Analysis of free vibrations

**Free undamped discrete vibration problem:**

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{0}$$

↓

**Ansatz:**

$$\mathbf{q}(t) = \alpha\mathbf{p}\cos(\omega t + \varphi)$$

↓

**Generalized eigenvalue problem:**

$$(\mathbf{K} - \omega^2\mathbf{M})\mathbf{p} = 0$$

**Solving the eigenvalue problem:**

$$n = \dim(\mathbf{K}) = \dim(\mathbf{M})$$

There are $n$ eigenvalues $\lambda_j$ and $n$ corresponding eigenvectors $\mathbf{p}_j$ such that:

**Eigenvalues** (*natural frequencies* squared): $\lambda_j = \omega_j^2$ are the roots of the characteristic polynomial:

$$\det(\mathbf{K} - \lambda\mathbf{M}) = 0.$$

**Eigenvectors** (*modal shapes*): $\mathbf{p}_j$ are the solution of the equation

$$(\mathbf{K} - \lambda_j\mathbf{M})\mathbf{p}_j = \mathbf{0}.$$

# Rigid body modes

In the semi-discrete weak form obtained via finite element discretization:

- The **mass matrix M** is symmetric and strictly positive definite.
- The **stiffness matrix K** is symmetric and positive semi-definite:

$$\mathbf{K}\mathbf{p} = \mathbf{0} \quad \text{for certain nonzero vectors } \mathbf{p}.$$

Consequently, the eigenvalues $\omega_j^2$ of the generalized eigenvalue problem are all *real* and *non-negative*:

$$0 \leq \omega_1 \leq \omega_2 \leq \cdots \leq \omega_n.$$

**Rigid body modes:** zero eigenvalues (i.e., $\omega_j = 0$) correspond to *rigid body motions*, where the system undergoes displacement without internal deformation.

# Orthonormalization of mode shapes

The eigenvectors (mode shapes) $\mathbf{p}_j$ are *orthonormal* with respect to the mass matrix $\mathbf{M}$ and the stiffness matrix $\mathbf{K}$.

Modal matrix $\mathbf{P} = \left[ \begin{array}{c:c:c:c} \mathbf{p}_1 & \mathbf{p}_2 & \dots & \mathbf{p}_n \end{array} \right]$ then

$$\mathbf{P}^T\mathbf{M}\mathbf{P} = \mathbf{I} \qquad \text{and} \qquad \mathbf{P}^T\mathbf{K}\mathbf{P} = \mathbf{\Lambda}$$

where $\mathbf{I}$ is the identity matrix of order $n$ and $\mathbf{\Lambda}$ the spectral matrix:

$$\mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \dots, \lambda_n) = \operatorname{diag}(\omega_1^2, \dots, \omega_n^2).$$

# Rayleigh's quotient

- Rayleigh's quotient of a generic vector $\mathbf{w}$:

$$\mathcal{R}(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{K} \mathbf{w}}{\mathbf{w}^T \mathbf{M} \mathbf{w}}$$

- For a given eigenvector $\mathbf{p}_i$, the Rayleigh quotient provides the corresponding eigenvalue $\omega_i^2$:

$$\mathcal{R}(\mathbf{p}_i) = \omega_i^2.$$

- The first variation of Rayleigh's quotient is zero at the eigenvectors:

$$\delta\mathcal{R}(\mathbf{w}) = 0 \quad \text{if and only if} \quad \mathbf{w} = \mathbf{p}_i,$$

  where $\mathbf{p}_i$ is an eigenvector.

- **Bounding theorem:** $\lambda_1 \leq \mathcal{R}(\mathbf{w}) \leq \lambda_n$

The Rayleigh quotient can be used to approximate the smallest eigenvalue $\omega_1^2$ by minimizing $\mathcal{R}(\mathbf{w})$ over all non-zero vectors $\mathbf{w}$.

# Inverse iteration algorithm

**Goal:** Compute **one** eigenvector using iterative refinement.

**Algorithm:**

1. Choose initial guess $\mathbf{p}^{(0)}$, spectral shift $\sigma$, tolerance $\varepsilon$.
2. Define shifted stiffness: $\mathbf{K}_\sigma = \mathbf{K} + \sigma\mathbf{M}$
3. For $k = 1, 2, \ldots$ until convergence:
   1. Solve: $\mathbf{K}_\sigma\, \bar{\mathbf{p}}^{(k)} = \mathbf{M}\, \mathbf{p}^{(k-1)}$
   2. Normalize: $\mathbf{p}^{(k)} = \bar{\mathbf{p}}^{(k)} / \sqrt{(\bar{\mathbf{p}}^{(k)})^T \mathbf{M} \bar{\mathbf{p}}^{(k)}}$
   3. Check: $\|\mathbf{p}^{(k)} - \mathbf{p}^{(k-1)}\| < \varepsilon$
4. Output: eigenvector $\mathbf{p}^{(k)}$, eigenvalue $\lambda^{(k)} = \mathcal{R}(\mathbf{p}^{(k)})$

**Key points:**

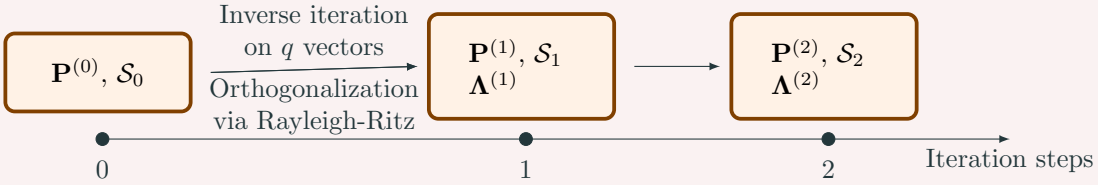
- Converges to eigenvector closest to $\sigma$ if $\mathbf{p}^{(0)}$ not $\mathbf{M}$-orthogonal to it.
- Without shift ($\sigma = 0$) $\rightarrow$ eigenvector of smallest eigenvalue.
- Convergence rate: $\rho = \frac{|\lambda_j - \sigma|}{|\lambda_{j+1} - \sigma|}$

# Subspace iteration method

**Goal:** Compute the lowest $n_{modes} \ll n$ eigenpairs $(\mathbf{p}_i, \lambda_i)$ of using a $q$-dimensional subspace iteration.

**Key ideas and advantages:**

- Iteration is performed on a subspace $\mathcal{S}_k = \text{span}(\mathbf{p}_1^{(k)}, \ldots, \mathbf{p}_q^{(k)})$, not individual vectors.
- At each iteration, inverse iteration is applied to all $q$ vectors in the subspace and then orthogonalized via Rayleigh-Ritz.
- Easier to initialize a subspace close to the span of required eigenvectors than individual vectors.



# Subspace iteration algorithm

**Inputs:** $\mathbf{K}, \mathbf{M}, n_{modes}, q > n_{modes}, \mathbf{P}^{(0)} \in \mathbb{R}^{n \times q}, \sigma, \varepsilon$

**Iteration:** For $k = 1, 2, \ldots$ until convergence:

1. Solve $\mathbf{K}_\sigma \overline{\mathbf{P}^{(k)}} = \mathbf{M}\mathbf{P}^{(k-1)}$ (inverse iteration on $q$ vectors)
2. Project and solve small eigenproblem:

$$\mathbf{K}^{(k)} = (\overline{\mathbf{P}^{(k)}})^T \mathbf{K} \overline{\mathbf{P}^{(k)}}, \quad \mathbf{M}^{(k)} = (\overline{\mathbf{P}^{(k)}})^T \mathbf{M} \overline{\mathbf{P}^{(k)}}$$

$$\mathbf{K}^{(k)}\mathbf{Z}^{(k)} = \mathbf{M}^{(k)}\mathbf{Z}^{(k)}\mathbf{\Lambda}^{(k)}$$

3. Update modal matrix: $\mathbf{P}^{(k)} = \overline{\mathbf{P}^{(k)}}\mathbf{Z}^{(k)}$, ensure $(\mathbf{P}^{(k)})^T \mathbf{M} \mathbf{P}^{(k)} = \mathbf{I}$
4. Check convergence: $\frac{|\lambda_i^{(k)} - \lambda_i^{(k-1)}|}{\lambda_i^{(k)}} < \varepsilon$

**Notes:**

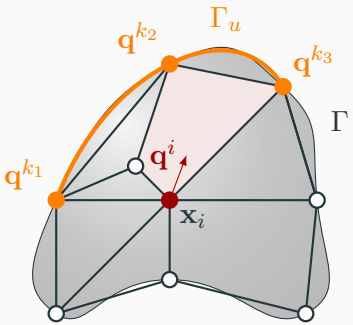
- Orthogonalization is critical to prevent collinearity of iteration vectors.
- Starting vectors should be linearly independent; larger $q$ increases convergence rate but also computational cost.
- After convergence, use Sturm sequence to verify the number of eigenvalues computed.

# Analysis of forced vibrations

# Forced vibrations of non-rotating conservative systems

The discretization of linear three-dimensional elastodynamics, as well as the analysis of vibrations in beams and plates via FEM, all lead to a system of ODE:

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{r}(t), \qquad \forall t \in [0, T]$$




- Generalized nodal displacements:

$$\mathbf{q}(t) = [\mathbf{q}^1(t), \ldots, \mathbf{q}^n(t)]^T.$$

- Excitation (force) vector: $\mathbf{r}(t) \neq \mathbf{0}$
- **Boundary conditions:** $\mathbf{q}^k = \hat{\mathbf{q}}^k$ for all $k$ such that $\mathbf{x}_k \in \Gamma_u$.
- **Initial conditions:** $\mathbf{q}(0) = \mathbf{u}_0$ and $\dot{\mathbf{q}}(0) = \mathbf{v}_0$

Interest in finding the **temporal response** $\mathbf{q}(t)$ of the structure.

# Modal basis




$$\mathbf{q}(t) = \mathbf{P}\mathbf{z}(t) = \sum_{i=1}^{n} \mathbf{p}_i z_i(t)$$

- $\mathbf{z}$: vector of modal coordinates
- $\mathbf{P}$: modal matrix
- $\mathbf{p}_i$: eigenvector of order $i$
- $z_i$: modal coordinate of order $i$

**Insertion of the change of basis in the forced regime equation:**

$$\mathbf{M}\mathbf{P}\ddot{\mathbf{z}}(t) + \mathbf{K}\mathbf{P}\mathbf{z}(t) = \mathbf{r}(t)$$
$$\mathbf{P}^{\mathrm{T}}\mathbf{M}\mathbf{P}\ddot{\mathbf{z}}(t) + \mathbf{P}^{\mathrm{T}}\mathbf{K}\mathbf{P}\mathbf{z}(t) = \mathbf{P}^{\mathrm{T}}\mathbf{r}(t)$$

# Decoupling of the forced regime

**Accounting for the orthogonality of eigenmodes**

$$\mathbf{P}^{\mathrm{T}}\mathbf{M}\mathbf{P} = \mathbf{I} \qquad \text{and} \qquad \mathbf{P}^{\mathrm{T}}\mathbf{K}\mathbf{P} = \mathbf{\Lambda}$$

- $\mathbf{I}$: identity matrix
- $\mathbf{\Lambda} = \mathrm{diag}(\omega_1^2, \ldots, \omega_i^2, \ldots, \omega_n^2)$: diagonal matrix of eigenvalues,

**Decoupling of the forced regime system:**

$$\mathbf{P}^{\mathrm{T}}\mathbf{M}\mathbf{P}\ddot{\mathbf{z}}(t) + \mathbf{P}^{\mathrm{T}}\mathbf{K}\mathbf{P}\mathbf{z}(t) = \mathbf{P}^{\mathrm{T}}\mathbf{r}(t)$$
$$\ddot{\mathbf{z}}(t) + \mathbf{\Lambda}\mathbf{z}(t) = \mathbf{s}(t)$$

- Projected initial conditions: $\mathbf{z}(0) = \mathbf{P}^{\mathrm{T}}\mathbf{M}\mathbf{u}_0$ and $\dot{\mathbf{z}}(0) = \mathbf{P}^{\mathrm{T}}\mathbf{M}\mathbf{v}_0$.
- $\mathbf{s}(t) = \mathbf{P}^{\mathrm{T}}\mathbf{r}(t)$: projection of $\mathbf{r}(t)$ onto the modal basis $\mathbf{P}$.

# Exact solution

**Component-wise form of the decoupled forced regime:** $(i = 1, 2, \ldots, n)$

$$\begin{aligned} \ddot{z}_i(t) + \omega_i^2 z_i(t) &= s_i(t) \\ z_i(0) &= \mathbf{p}_i^{\mathrm{T}} \mathbf{M} \mathbf{u}_0 \\ \dot{z}_i(0) &= \mathbf{p}_i^{\mathrm{T}} \mathbf{M} \mathbf{v}_0 \end{aligned}$$

Exact solution by Laplace transform (convolution or Duhamel's integral):

$$z_i(t) = z_i(0) \cos(\omega_i t) + \frac{\dot{z}_i(0)}{\omega_i} \sin(\omega_i t) + \frac{1}{\omega_i} \int_0^t \sin\big(\omega_i (t - \tau)\big)\, s_i(\tau)\, \mathrm{d}\tau$$

$$\begin{aligned} \mathbf{q}(t) = \sum_{i=1}^{n} \mathbf{p}_i z_i(t) = \sum_{i=1}^{n} \mathbf{p}_i \left( \mathbf{p}_i^{\mathrm{T}} \mathbf{M} \mathbf{u}_0 \cos(\omega_i t) + \frac{1}{\omega_i} \mathbf{p}_i^{\mathrm{T}} \mathbf{M} \mathbf{v}_0 \sin(\omega_i t) \right) \\ + \sum_{i=1}^{n} \mathbf{p}_i \left( \frac{1}{\omega_i} \int_0^t s_i(t - \tau) \sin(\omega_i \tau)\, \mathrm{d}\tau \right) \end{aligned}$$

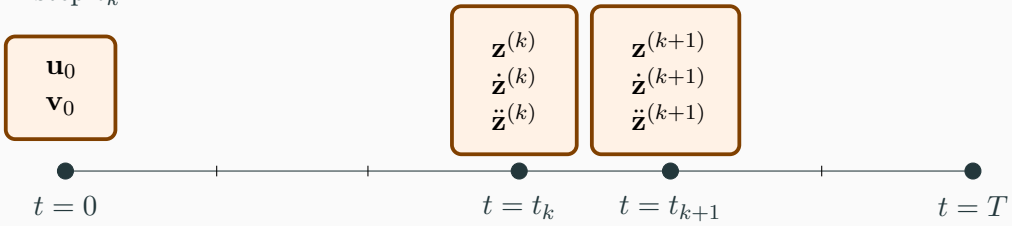
# Direct integration methods

# Finite difference method for time integration

- Solve dynamic equilibrium equations by replacing time derivatives with discrete approximations.
- Time domain $[0, T]$ is discretized into $N_t$ equal intervals $\Delta t = T/N_t$:

$$t_k = k\Delta t \qquad k = 0, \dots N_t.$$

- Displacement, velocity, and acceleration are approximated within each time step $t_k$.



# Approximate solution via the finite difference method

**General kinematic scheme (for second-order differential equation):**

$$\begin{bmatrix} z_i^{(k)} \\ \dot{z}_i^{(k)} \end{bmatrix} = \sum_{j=1}^{q} \alpha_j \begin{bmatrix} z_i^{(k-j)} \\ \dot{z}_i^{(k-j)} \end{bmatrix} + \Delta t \sum_{j=0}^{q} \beta_j \begin{bmatrix} \dot{z}_i^{(k-j)} \\ \ddot{z}_i^{(k-j)} \end{bmatrix}, \quad k = 1, 2, \ldots; \quad k \geq q$$

- $z_i^{(k)}, \dot{z}_i^{(k)}, \ddot{z}_i^{(k)}$: component $i$ of displacement, velocity, and acceleration at time step $k$,
- $q$, $\alpha_j$, $\beta_j$ for $j = 1, \ldots, q$: given constants,
- $\Delta t = t_k - t_{k-1}$: time step,
- $k$: time step index.

# Types of time integration schemes

**Single-step methods** ($q = 1$):

- Use only the state at the previous time step to compute the current time step.
- Widely used due to balance between accuracy and efficiency.
- *Example:* Newmark methods are commonly used in finite element software.

**Multi-step methods** ($q > 1$):

- Use multiple previous time steps to compute the current time step.
- *Examples:* Houbolt's method, Wilson-$\theta$ method, Park's algorithm,

**Explicit vs implicit** ($\beta_0$):

- Explicit: $\beta_0 = 0$, current values are computed directly from known past values.
- Implicit: $\beta_0 \neq 0$, require solving equations at each time step.

# Classification of finite difference methods

**General kinematic scheme:**

$$\begin{bmatrix} z_i^{(k)} \\ \dot{z}_i^{(k)} \end{bmatrix} = \sum_{j=1}^{q} \alpha_j \begin{bmatrix} z_i^{(k-j)} \\ \dot{z}_i^{(k-j)} \end{bmatrix} + \Delta t \sum_{j=0}^{q} \beta_j \begin{bmatrix} \dot{z}_i^{(k-j)} \\ \ddot{z}_i^{(k-j)} \end{bmatrix}, \quad k = 1, 2, \ldots; \quad k \geq q$$

- $q = 1$ : one-step scheme $\rightarrow$ Newmark method(s)
  - $\beta_0 = 0$ : *explicit scheme* $\quad z_i^{(k)} \leftarrow z_i^{(k-1)}, \dot{z}_i^{(k-1)}$
  - $\beta_0 \neq 0$ : *implicit scheme* $\quad z_i^{(k)} \leftarrow z_i^{(k-1)}, \dot{z}_i^{(k-1)}, \ddot{z}_i^{(k)}$
- $q > 1$ : multi-step scheme $\rightarrow$ Park, Houbolt, Wilson, ...
  - $\beta_0 = 0$ : *explicit scheme*
  - $\beta_0 \neq 0$ : *implicit scheme*

# Newmark's one-step method ($t = 0$)

**Resolution of dynamic equations using a one-step Newmark's methods:**

- Initial conditions:

$$z_i^{(0)} = \mathbf{p}_i^{\mathrm{T}} \mathbf{M} \mathbf{u}_0$$
$$\dot{z}_i^{(0)} = \mathbf{p}_i^{\mathrm{T}} \mathbf{M} \mathbf{v}_0$$

- Dynamic equation:

$$\ddot{z}_i^{(k)} + \omega_i^2 z_i^{(k)} = s_i^{(k)}$$

**Computation of initial acceleration:**

$$\ddot{z}_i^{(0)} = -\omega_i^2 z_i^{(0)} + s_i^{(0)}$$

# Newmark's one-step method ($t = t_{k-1}$ to $t = t_k$)

**Kinematic scheme for one-step Newmark's methods:**

$$\begin{bmatrix} z_i^{(k)} \\ \dot{z}_i^{(k)} \end{bmatrix} = \alpha_1 \begin{bmatrix} z_i^{(k-1)} \\ \dot{z}_i^{(k-1)} \end{bmatrix} + \Delta t \left( \beta_0 \begin{bmatrix} \dot{z}_i^{(k)} \\ \ddot{z}_i^{(k)} \end{bmatrix} + \beta_1 \begin{bmatrix} \dot{z}_i^{(k-1)} \\ \ddot{z}_i^{(k-1)} \end{bmatrix} \right) \qquad (k = 1, 2, \ldots)$$

- $\alpha_1$, $\beta_0$, and $\beta_1$: parameters characterizing the different variants of the schemes.
- To compute $\alpha_1$, $\beta_0$, and $\beta_1$ we use Taylor expansion with integral remainder:

$$\underbrace{z_i(t_k)}_{z_i^{(k)}} = \underbrace{z_i(t_{k-1})}_{z_i^{(k-1)}} + \Delta t \, \underbrace{\dot{z}_i(t_{k-1})}_{\dot{z}_i^{(k-1)}} + \int_{t_{k-1}}^{t_k} (t_k - \tau) \ddot{z}_i(\tau) \, d\tau$$

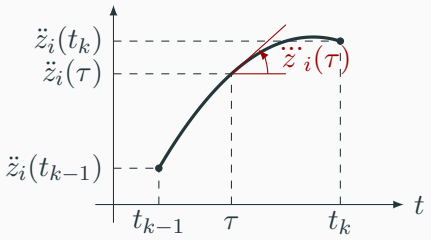
$$\underbrace{\dot{z}_i(t_k)}_{\dot{z}_i^{(k)}} = \underbrace{\dot{z}_i(t_{k-1})}_{\dot{z}_i^{(k-1)}} + \int_{t_{k-1}}^{t_k} \ddot{z}_i(\tau) \, d\tau$$

# First-order integral remainder approximation

**Taylor series for acceleration:**
let $\beta \in ]0, 1/2[$ a constant, then

$$\times (1-2\beta) \qquad \ddot{z}_i(t_{k-1}) = \ddot{z}_i(\tau) - (\tau - t_{k-1})\,\dddot{z}_i(\tau) + \dots$$
$$\times 2\beta \qquad \ddot{z}_i(t_k) = \ddot{z}_i(\tau) + (t_k - \tau)\,\dddot{z}_i(\tau) + \dots$$

$$\Rightarrow (1-2\beta)\ddot{z}_i(t_{k-1}) + 2\beta\ddot{z}_i(t_k) \approx \ddot{z}_i(\tau)$$




Notice that terms in $\dddot{z}_i(\tau)$ and higher are neglected because they are multiplied in displacement by $\Delta t^3$, and in velocity by $\Delta t^2$.

# Taylor formula with integral remainder for modal displacements

Integral quadrature in Taylor expansion:

$$\begin{aligned}\int_{t_{k-1}}^{t_k} (t_k - \tau)\ddot{z}_i(\tau)\, d\tau &= \int_{t_{k-1}}^{t_k} (t_k - \tau)\left[(1-2\beta)\ddot{z}_i^{(k-1)} + 2\beta\ddot{z}_i^{(k)}\right] d\tau \\ &= \dots \\ &= \Delta t^2 \left[\left(\frac{1}{2} - \beta\right)\ddot{z}_i^{(k-1)} + \beta\ddot{z}_i^{(k)}\right]\end{aligned}$$

**Kinematic scheme for displacements:**

$$z_i^{(k)} = z_i^{(k-1)} + \Delta t \dot{z}_i^{(k-1)} + \Delta t^2 \left[\left(\frac{1}{2} - \beta\right)\ddot{z}_i^{(k-1)}\right] + \beta \Delta t^2 \ddot{z}_i^{(k)}$$

# First-order integral remainder approximation

**Taylor series for acceleration:** let $\gamma \in ]0,1[$ a second constant, then

$$\Rightarrow (1-\gamma)\ddot{z}_i(t_{k-1}) + \gamma\ddot{z}_i(t_k) \approx \ddot{z}_i(\tau)$$

**Integral quadrature in Taylor expansion:**

$$\int_{t_{k-1}}^{t_k} \ddot{z}_i(\tau)\, d\tau = \int_{t_{k-1}}^{t_k} \left[(1-\gamma)\ddot{z}_i^{(k-1)} + \gamma\ddot{z}_i^{(k)}\right] d\tau = \Delta t \left[(1-\gamma)\,\ddot{z}_i^{(k-1)} + \gamma\ddot{z}_i^{(k)}\right]$$

**Kinematic scheme for velocity:**

$$\dot{z}_i^{(k)} = \dot{z}_i^{(k-1)} + \Delta t \left[(1-\gamma)\,\ddot{z}_i^{(k-1)}\right] + \gamma\Delta t\ddot{z}_i^{(k)}$$

# Newmark's one-step method ($t = t_{k-1}$ to $t = t_k$)

Stability control parameter $\beta$: $0 \leq \beta \leq 1/4$

Numerical dissipation parameter $\gamma$: $1/2 \leq \gamma \leq 3/4$.

- **Kinematic scheme for displacement:**

$$z_i^{(k)} = z_i^{(k-1)} + \Delta t \dot{z}_i^{(k-1)} + \Delta t^2 \left[ \left( \frac{1}{2} - \beta \right) \ddot{z}_i^{(k-1)} \right] + \beta \Delta t^2 \ddot{z}_i^{(k)}$$

- **Kinematic scheme for velocity:**

$$\dot{z}_i^{(k)} = \dot{z}_i^{(k-1)} + \Delta t \left[ (1 - \gamma) \ddot{z}_i^{(k-1)} \right] + \gamma \Delta t \ddot{z}_i^{(k)}$$

- **Dynamic equation:**

$$\ddot{z}_i^{(k)} + \omega_i^2 z_i^{(k)} = s_i^{(k)}$$

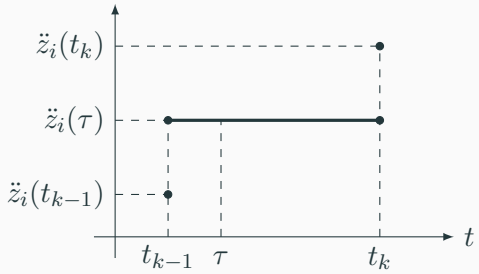
# Average acceleration method (Trapezoidal rule)

**Special case:**

$$\beta = \frac{1}{4} \quad \text{and} \quad \gamma = \frac{1}{2}$$

Then for $t_{k-1} \leq \tau \leq t_k$ we have

$$\ddot{z}_i(\tau) = \frac{\ddot{z}_i(t_{k-1}) + \ddot{z}_i(t_k)}{2}$$




**Kinematic schemes for displacement and velocity:**

$$z_i^{(k)} = z_i^{(k-1)} + \Delta t \dot{z}_i^{(k-1)} + \frac{1}{4}\Delta t^2 \left(\ddot{z}_i^{(k-1)} + \ddot{z}_i^{(k)}\right)$$

$$\dot{z}_i^{(k)} = \dot{z}_i^{(k-1)} + \frac{1}{2}\Delta t \left(\ddot{z}_i^{(k-1)} + \ddot{z}_i^{(k)}\right)$$




*"The person who learns the most in any classroom is the teacher."*

*— James Clear*

**Please complete the in-depth evaluation survey!**